Notes on Gauge Theory

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2 Principal Bundles and Connections

2.1 Principal Bundles

Principal bundles are both a special type of fiber bundle and the most important one. In the following, we will introduce the definition of principal bundles and some basic structures on them.

Definition 2.1 (G-action)**.** A **smooth right action** of a Lie group G on a manifold P is a smooth map

$$\mu : M \times G \to M,$$

denoted by $\mu(x, g) := x \cdot g$, such that for all $x \in M$ and $g, h \in G$,

$$x \cdot e = x, \quad (x \cdot g) \cdot h = x \cdot (gh).$$

where e is the identity element of G. A manifold M with a smooth right G-action is called a (**right**) G**-manifold**.

We say the action is **free** if for all $x \in M$, $x \cdot g = x$ implies $g = e$.

Definition 2.2 (G-equivariant map)**.** Let M and N be two G-manifolds. A smooth map $f : M \to N$ is called G**-equivariant** if for all $x \in M$ and $g \in G$,

$$f(x \cdot g) = f(x) \cdot g.$$

Definition 2.3 (Principal G-bundle)**.** A **principal** G**-bundle** is a fiber bundle $\pi : P \to M$ with fiber G together with a free right G-action on P such that for each $x \in M$, the local trivialization

$$\phi_U : \pi^{-1}(U) \to U \times G,$$

is G-equivariant, where the right G-action on $U \times G$ is given by

$$(x, h) \cdot g = (x, hg), \quad \forall x \in U, g, h \in G.$$

By a **local trivialization** we mean a open neighborhood U of x and a diffeomorphism $\phi_U : \pi^{-1}(U) \to U \times G$ satisfying the following commutative diagram:

$$\begin{tikzcd}
\pi^{-1}(U) \arrow[rr, "\phi_U"] \arrow[dr, "\pi"'] & & U \times G \arrow[dl, "\mathrm{pr}_1"] \\
& U &
\end{tikzcd}$$

where $p_1 : U \times G \to U$ is the projection map.

Example 2.4. Let G be a Lie group and M be a smooth manifold, then the trivial bundle $p_1 : M \times G \to M$ is a principal G-bundle over M.

Example 2.5. Let G be a Lie group and H be a closed subgroup of G. Then the quotient space G/H is a smooth manifold and the projection map $\pi : G \to G/H$ is a principal H-bundle over G/H.

Example 2.6. Let $\pi : E \to M$ be a rank n vector bundle over M. Then the frame bundle $\mathrm{Fr}(E)$ is a principal $\mathrm{GL}(n, \mathbb{R})$-bundle over M.

Proposition 2.7. *Let $\pi : P \to M$ be a principal G-bundle. Then the right G-action on P is transitive on each fiber.*

Proof. Since G acts transitively on $\{x\} \times G$ and the fiber diffeomorphism $\phi_{U,x} : P_x \to G$ is G-equivariant, thus G acts transitively on P_x. □

Lemma 2.8. *For any group G, all right G-equivariant maps are exactly the left translations.*

Proof. Let $f : G \to G$ be a right G-equivariant map, then for all $g \in G$,

$$f(g) = f(e \cdot g) = f(e) \cdot g = l_{f(e)}(g),$$

where $l_{f(e)} : G \to G$ is the left translation by $f(e)$. □

Definition 2.9 (Cocycle definition)**.** A **principal G-bundle** is a fiber bundle $\pi : P \to M$ with fiber G together with a local trivialization $\{(U_\alpha, \phi_\alpha)\}$ such that for each $U_{\alpha\beta} := U_\alpha \cap U_\beta \neq \emptyset$, there exists a smooth map $g_{\alpha\beta} : U_{\alpha\beta} \to G$ such that

$$\phi_\alpha \circ \phi_\beta^{-1}(x, h) = (x, g_{\alpha\beta}(x)h), \quad \forall (x, h) \in U_{\alpha\beta} \times G.$$

In addition, the transition functions $\{g_{\alpha\beta}\}$ satisfy the following cocycle condition:

$$\begin{gathered} g_{\alpha\alpha}(x) = e, \quad \forall x \in U_\alpha, \\ g_{\alpha\beta}(x) g_{\beta\gamma}(x) = g_{\alpha\gamma}(x), \quad \forall x \in U_\alpha \cap U_\beta \cap U_\gamma \neq \emptyset. \end{gathered}$$

Remark. By Lemma 2.8, one can easily check that the two definitions of principal bundles are equivalent. These two definitions characterize G-principal bundles from different perspectives.

2.1.1 Fundamental Vector Fields

Suppose P is a G-manifold, then for each $A \in \mathfrak{g}$, we can define a vector field $\underline{A}$ on P by

$$\underline{A}_p := \left.\frac{d}{dt}\right|_{t=0} p \cdot \exp(tA), \quad \forall p \in P.$$

The vector field $\underline{A}$ is called the **fundamental vector field** associated to A.

Fact 2.1. For each $A \in \mathfrak{g}$, $\underline{A}$ is a C^∞ vector field on P.

The fundamental vector field construction gives rise to a map

$$\sigma : \mathfrak{g} \to \mathfrak{X}(P),^{1} \quad \sigma(A) := \underline{A}.$$

For $p \in P$, define $j_p : G \to P$ by $j_p(g) := p \cdot g$, then $j_{p*} := j_{p*e}$ is

$$j_{p*}(A) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} j_p(\exp(tA)) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} p \cdot \exp(tA) = \underline{A}_p.$$

This gives another discription of the fundamental vector field: $\underline{A}_p = j_{p*}(A)$. It also implies that the map $\sigma : \mathfrak{g} \to \mathfrak{X}(P)$ is $\mathbb{R}$-linear. In fact, we have

[1] $\mathfrak{X}(P) := C^\infty(P, TP)$ is the space of smooth vector fields on P.

Fact 2.2. The map $\sigma : \mathfrak{g} \to \mathfrak{X}(P)$ is a Lie algebra homomorphism, i.e.,

$$[\underline{A}, \underline{B}] = \underline{[A, B]}, \quad \forall A, B \in \mathfrak{g}.$$

Example 2.10. Let G acts on itself by right translation, then the fundamental vector field $\underline{A}$ is exactly the left-invariant vector field generated by A.

For $g \in G$, let $c_g : G \to G$ be the conjugation map $c_g(h) := ghg^{-1}$. Then the differential of c_g at the identity element e is the adjoint representation $\mathrm{Ad}_g := (c_g)_* : \mathfrak{g} \to \mathfrak{g}$.

Proposition 2.11. *For each $A \in \mathfrak{g}$ and $g \in G$, we have*

$$(r_g)_*\underline{A} = \underline{\mathrm{Ad}_{g^{-1}}(A)}.$$

where $r_g : P \to P$ is the right translation by g, i.e., $r_g(p) := p \cdot g$.

Proof. For each $p \in P$ and $h \in G$, we have

$$(r_g \circ j_p)(h) = r_g(p \cdot h) = (p \cdot h) \cdot g = pg \cdot (g^{-1}hg) = (j_{pg} \circ c_{g^{-1}})(h).$$

By the chain rule,

$$(r_g)_*(\underline{A}_p) = (r_g)_* j_{p*}(A) = j_{pg*}(c_{g^{-1}})_*(A) = j_{pg*}(\mathrm{Ad}_{g^{-1}}(A)) = \underline{\mathrm{Ad}_{g^{-1}}(A)}_{pg}.$$ □

Fact 2.3. Let P be a G-manifold, then the fundamental vector fields $\underline{A}$ vanishes at p if and only if $A \in \mathrm{Lie}(\mathrm{Stab}(p))$. Thus for any $p \in P$, the kernel of $j_{p*} : \mathfrak{g} \to T_pP$ is exactly $\mathrm{Lie}(\mathrm{Stab}(p))$. In particular, if the G-action on P is free, then j_{p*} is injective for all $p \in P$.

2.1.2 Vertical Subbundle of TP

Let $\pi : P \to M$ be a principal G-bundle, at each $p \in P$, the differential $\pi_{*p} : T_pP \to T_{\pi(p)}M$ is surjective. We define $\mathcal{V}_p := \ker \pi_{*p} \subset T_pP$ to be the **vertical tangent space** at p. Then $\mathcal{V} := \bigsqcup_{p\in P} \mathcal{V}_p$ is a subbundle of TP, called the **vertical subbundle** of TP. An element of $\mathcal{V}_p$ is called a **vertical tangent vector** at p.

We have a short exact sequence of vector spaces

$$0 \longrightarrow \mathcal{V}_p \hookrightarrow T_pP \xrightarrow{\pi_{*p}} T_{\pi(p)}M \longrightarrow 0$$

and

$$\dim \mathcal{V}_p = \dim T_pP - \dim T_{\pi(p)}M = \dim G.$$

Proposition 2.12. *For any $A \in \mathfrak{g}$, the fundamental vector field $\underline{A}$ is vertical at every point $p \in P$.*

Proof. Since $(\pi \circ j_p)(g) = \pi(p \cdot g) = \pi(p)$ is a constant map for all $g \in G$, we have $\pi_{*p}(\underline{A}_p) = \pi_{*p} j_{p*}(A) = (\pi \circ j_p)_*(A) = 0$. □

This proposition implies that $j_{p*}(\mathfrak{g}) \subset \mathcal{V}_p$ for all $p \in P$. In fact, we have the following stronger result.

Proposition 2.13. *For each $p \in P$, the map $j_{p*} : \mathfrak{g} \to \mathcal{V}_p$ is an isomorphism.*

Proof. Since G acts freely on P, the map j_{p*} is injective. On the other hand, $\dim \mathcal{V}_p = \dim G = \dim \mathfrak{g}$, thus j_{p*} is an isomorphism. $\square$

Corollary 2.14. *The vertical vectors at $p \in P$ are exactly the fundamental vector at p. Moreover, the vertical subbundle $\mathcal{V}$ of TP is isomorphic to the trivial bundle $P \times \mathfrak{g}$.*

Proof. The first statement follows from the previous proposition. For the second statement, let $A_1, \dots, A_l$ be a basis of $\mathfrak{g}$, where $l = \dim G$, then $\underline{A}_1, \dots, \underline{A}_l$ is a global frame of $\mathcal{V}$, thus $\mathcal{V}$ is a trivial bundle. $\square$

The map $\pi_* : TP \to TM$ induces a bundle map $\tilde{\pi}_* : TP \to \pi^*TM$ over P. Then we have a short exact sequence of vector bundles over P:

$$0 \longrightarrow \mathcal{V} \hookrightarrow TP \xrightarrow{\tilde{\pi}_*} \pi^*TM \longrightarrow 0$$

2.2 Connections on Principal Bundles

2.2.1 Horizontal Distribution

In the category of smooth vector bundles, all short exact sequence of vector bundles splits. Corresponding to the short exact sequence appearing in the previous chapter, there exists a smooth subbundle $\mathcal{H}$ of TP such that $TP = \mathcal{V} \oplus \mathcal{H}$. We call $\mathcal{H}$ a **horizontal distribution** of TP.

In general, there is no canonical choice of $\mathcal{H}$. A choice of $\mathcal{H}$ is the same as a splitting $k : \pi^*TM \to TP$ such that $\tilde{\pi}_* \circ k = \mathrm{id}_{\pi^*TM}$.

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{V} \arrow[r, hook] & TP \arrow[r, shift left, "\tilde{\pi}_*"] & \pi^*TM \arrow[l, shift left, "k"] \arrow[r] & 0 \end{tikzcd}$$

We say $\mathcal{H}$ is **right-invariant** if for all $g \in G$ and $p \in P$, $r_{g*p}(\mathcal{H}_p) = \mathcal{H}_{pg}$, where $r_g : P \to P$, $r_g(p) = p \cdot g$.

2.2.2 Ehresmann Connection

Let $\pi : P \to M$ be a principal G-bundle, $\mathcal{H}$ be a horizontal distribution of TP. For each $p \in P$, we define a projection map $v : T_pP = \mathcal{V}_p \oplus \mathcal{H}_p \to \mathcal{V}_p$. For $Y_p \in T_pP$, we call $v(Y_p)$ the **vertical component** of Y_p.[1]

Let $\iota : \mathcal{V} \hookrightarrow TP$ be the inclusion map, then $v \circ \iota = \mathrm{id}_{\mathcal{V}}$. We actually obtain another splitting of the short exact sequence:

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{V} \arrow[r, hook, shift left, "\iota"] & TP \arrow[l, shift left, "v"] \arrow[r, "\tilde{\pi}_*"] & \pi^*TM \arrow[r] & 0 \end{tikzcd}$$

Recall that for each $p \in P$, the map $j_p : G \to P$ defined by $j_p(g) = p \cdot g$. We define a $\mathfrak{g}$-valued 1-form ω on P by

$$\omega_p := j_{p*}^{-1} \circ v : T_pP \xrightarrow{v} \mathcal{V}_p \xrightarrow{j_{p*}^{-1}} \mathfrak{g}.$$

[1]Although $\mathcal{V}_p$ is intrinsically defined, the vertical component of a tangent vector depends on the choice of $\mathcal{H}_p$.

In terms of ω, the vertical component of $Y_p \in T_pP$ is given by

$$v(Y_p) = j_{p*}(\omega_p(Y_p)).$$

In summary, given a projection $v : TP \to \mathcal{V}$, we induce the following composition of maps:

$$\omega : TP \xrightarrow{v} \mathcal{V} \xrightarrow{j_*^{-1}} P \times \mathfrak{g} \xrightarrow{\mathrm{pr}_2} \mathfrak{g} \quad \rightsquigarrow \quad \omega \in \Omega^1(P, \mathfrak{g}).$$

So the information of the horizontal distribution $\mathcal{H}$ is encoded in ω.

Proposition 2.15. *If $\mathcal{H}$ is right-invariant, then the $\mathfrak{g}$-valued 1-form ω defined above satisfies the following properties:*

1. $\omega_p(\underline{A}_p) = A$, $\forall\, p \in P$, $A \in \mathfrak{g}$.
2. *(G-equivalence)* $(r_g)^*\omega = (\mathrm{Ad}_{g^{-1}})\omega$, $\forall\, g \in G$.
3. *ω is smooth.*

We say a $\mathfrak{g}$-valued 1-form ω on P satisfying the above properties is an ***Ehresmann connection*** *or simply a* ***connection*** *on P.*

Proof. *1.* Since $\underline{A}$ is already vertical,

$$\omega_p(\underline{A}_p) = j_{p*}^{-1}(v(\underline{A}_p)) = j_{p*}^{-1}(\underline{A}_p) = A.$$

2. For each $p \in P$ and $Y_p \in T_pP$, we need to show

$$\omega_{pg}((r_g)_*Y_p) = (\mathrm{Ad}_{g^{-1}})\omega_p(Y_p).$$

Since both side are linear in Y_p, we only need to check the above equality for $Y_p \in \mathcal{V}_p$ and $Y_p \in \mathcal{H}_p$ respectively.

If $Y_p \in \mathcal{V}_p$, then $Y_p = \underline{A}_p$ for some $A \in \mathfrak{g}$, thus

$$\omega_{pg}(r_{g*}\underline{A}_p) = \omega_{pg}(\underline{\mathrm{Ad}_{g^{-1}}(A)}_{pg}) = \mathrm{Ad}_{g^{-1}}(A) = (\mathrm{Ad}_{g^{-1}})\omega_p(\underline{A}_p).$$

If $Y_p \in \mathcal{H}_p$, by the right-invariance of $\mathcal{H}$, we have $r_{g*}(Y_p) \in \mathcal{H}_{pg}$, thus

$$\omega_{pg}(r_{g*}Y_p) = 0 = (\mathrm{Ad}_{g^{-1}})\omega_p(Y_p).$$

3. Omit. □

Viewing ω as a map $\omega : TP \to \mathfrak{g}$. Both TP and $\mathfrak{g}$ are G-manifolds, where

$$g \cdot Y_p := (r_g)_*Y_p, \quad g \cdot A := \mathrm{Ad}_{g^{-1}}(A).$$

Then the G-equivalence condition of ω

$$\omega_{pg}((r_g)_*Y_p) = (\mathrm{Ad}_{g^{-1}})\omega_p(Y_p),$$

can be rephrased as $\omega : TP \to \mathfrak{g}$ is a G-equivariant map.

2.2.3 Curvature of Connections